D Y PATIL AGRICULTURE & TECHNICAL UNIVERSITY TALSANDE

MCA SANDWICH

PROJECT NAME

Heart Disease Analysis

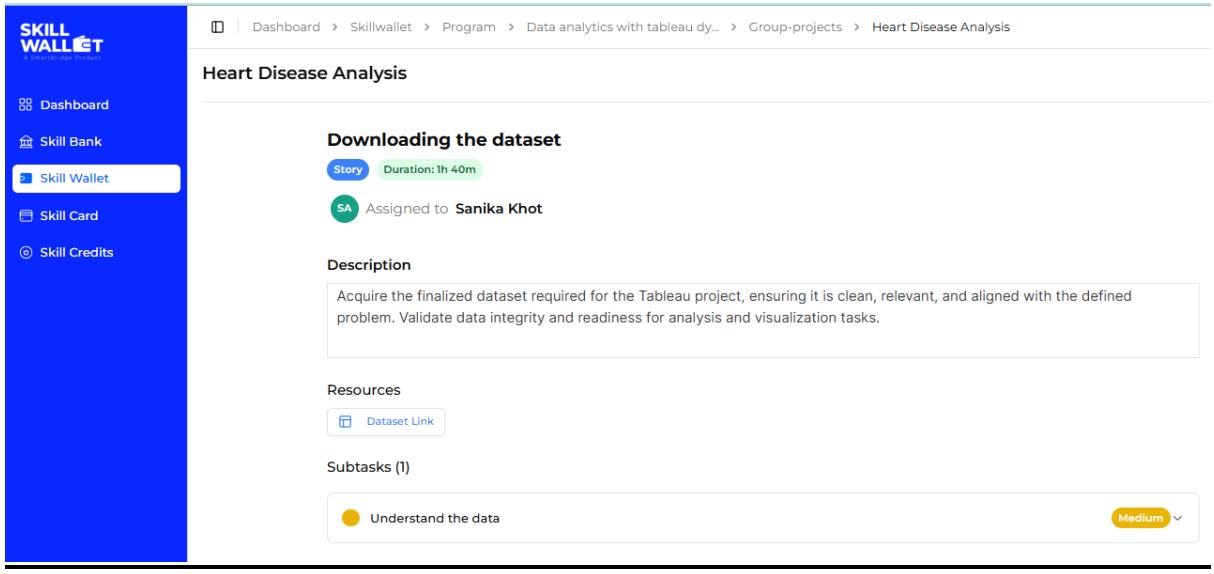
TEAM MEMBERS

1) Shweta Ramesh Shinde
2) Sanika Vishnu Khot
3) Rohan Raghunath Kharat
4) Pritam Shashikant Yadav

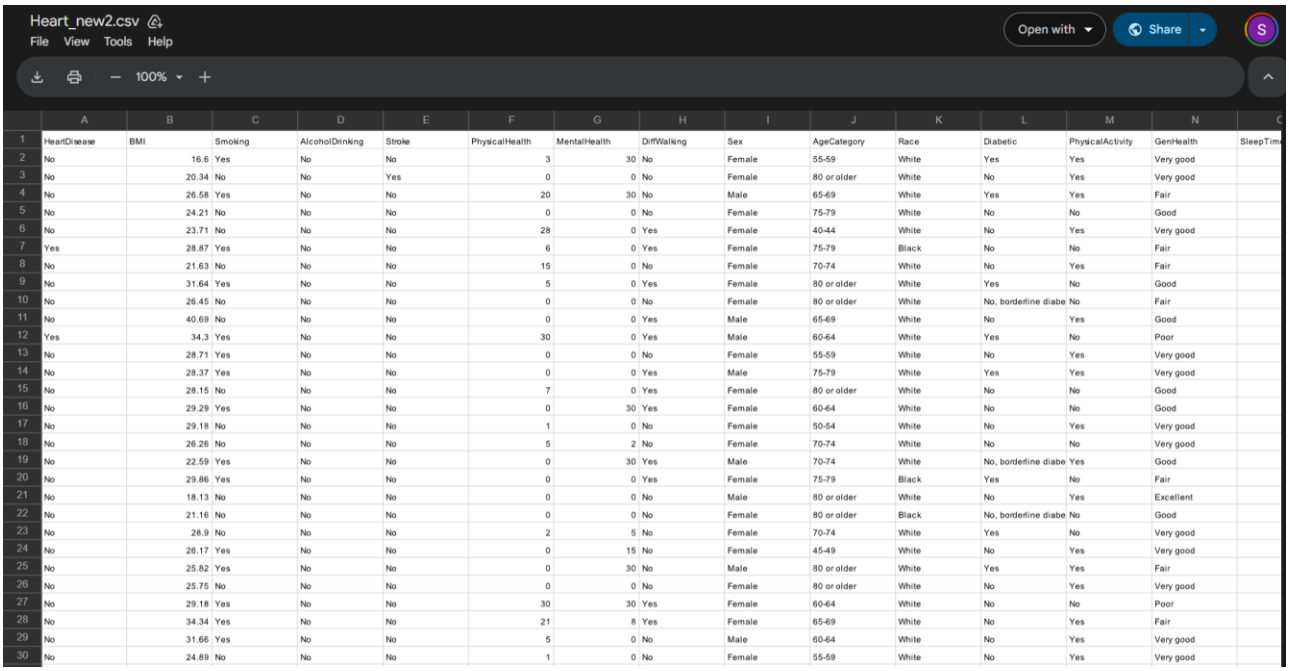
1) Data Collection & Extraction from Database:

❖ Collect the dataset:

1) Data is Collected from Dataset link Which is provided already so click on **Dataset Link**.

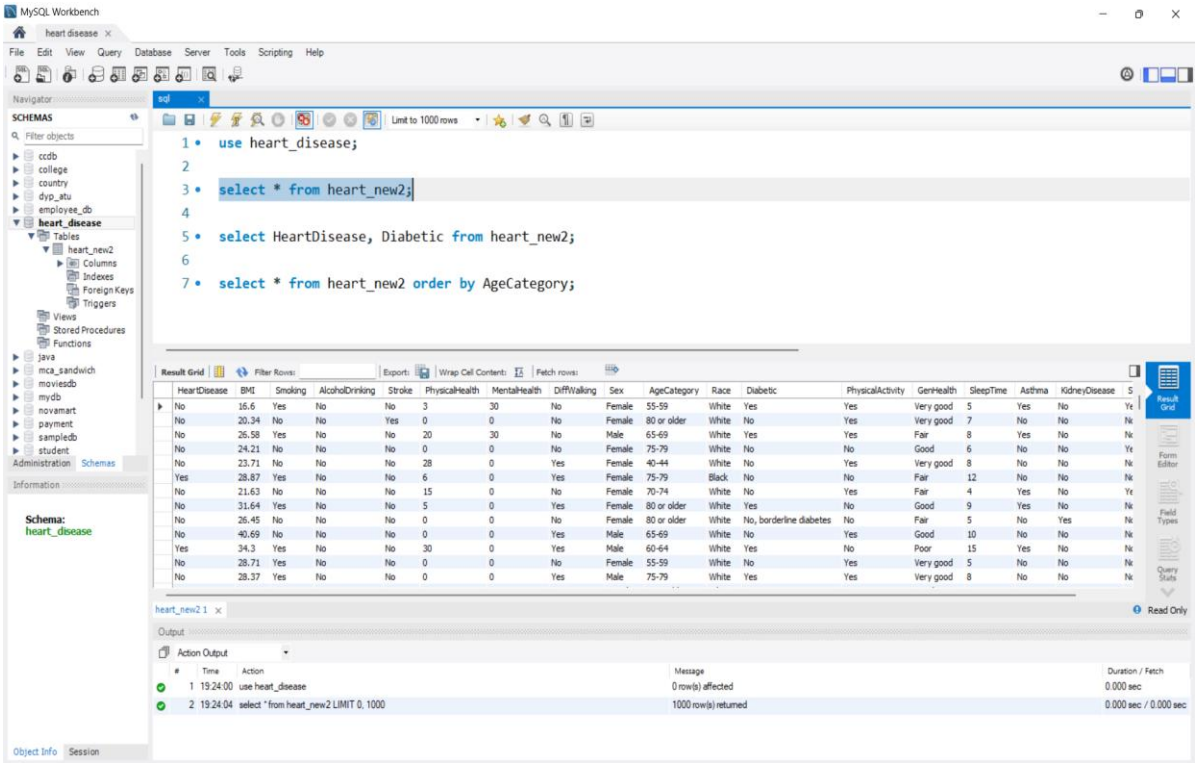


2) From the Google Drive Download the Dataset **Heart_new2.csv** by clicking on **Download** button,



	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	HeartDisease	BMI	Smoking	AlcoholDrinking	Stroke	PhysicalHealth	MentalHealth	DiffWalking	Sex	AgeCategory	Race	Diabetic	PhysicalActivity	GenHealth
2	No	16.6	Yes	No	No	3	30	No	Female	55-59	White	Yes	Yes	Very good
3	No	20.34	No	No	Yes	0	0	No	Female	80 or older	White	No	Yes	Very good
4	No	26.58	Yes	No	No	20	30	No	Male	65-69	White	Yes	Yes	Fair
5	No	24.21	No	No	No	0	0	No	Female	75-79	White	No	No	Good
6	No	23.71	No	No	No	28	0	Yes	Female	40-44	White	No	Yes	Very good
7	Yes	28.87	Yes	No	No	6	0	Yes	Female	75-79	Black	No	No	Fair
8	No	21.63	No	No	No	15	0	No	Female	70-74	White	No	Yes	Fair
9	No	31.64	Yes	No	No	5	0	Yes	Female	80 or older	White	Yes	No	Good
10	No	26.45	No	No	No	0	0	No	Female	80 or older	White	No, borderline diabe	No	Fair
11	No	40.69	No	No	No	0	0	Yes	Male	65-69	White	No	Yes	Good
12	Yes	34.3	Yes	No	No	30	0	Yes	Male	60-64	White	Yes	No	Poor
13	No	28.71	Yes	No	No	0	0	No	Female	55-59	White	No	Yes	Very good
14	No	28.37	Yes	No	No	0	0	Yes	Male	75-79	White	Yes	Yes	Very good
15	No	28.15	No	No	No	7	0	Yes	Female	80 or older	White	No	No	Good
16	No	29.29	Yes	No	No	0	30	Yes	Female	60-64	White	No	No	Good
17	No	29.18	No	No	No	1	0	No	Female	50-54	White	No	Yes	Very good
18	No	26.26	No	No	No	5	2	No	Female	70-74	White	No	No	Very good
19	No	22.59	Yes	No	No	0	30	Yes	Male	70-74	White	No, borderline diabe	Yes	Good
20	No	29.86	Yes	No	No	0	0	Yes	Female	75-79	Black	Yes	No	Fair
21	No	18.13	No	No	No	0	0	No	Male	80 or older	White	No	Yes	Excellent
22	No	21.16	No	No	No	0	0	No	Female	80 or older	Black	No, borderline diabe	No	Good
23	No	28.9	No	No	No	2	5	No	Female	70-74	White	Yes	No	Very good
24	No	26.17	Yes	No	No	0	15	No	Female	45-49	White	No	Yes	Very good
25	No	25.82	Yes	No	No	0	30	No	Male	80 or older	White	Yes	Yes	Fair
26	No	25.75	No	No	No	0	0	No	Female	80 or older	White	No	Yes	Very good
27	No	29.18	Yes	No	No	30	30	Yes	Female	60-64	White	No	No	Poor
28	No	34.34	Yes	No	No	21	8	Yes	Female	65-69	White	No	Yes	Fair
29	No	31.66	Yes	No	No	5	0	No	Male	60-64	White	No	Yes	Very good
30	No	24.89	No	No	No	1	0	No	Female	55-59	White	No	Yes	Very good

❖ Storing Data in DB & Perform SQL Operations medium



❖ Loading the Dataset:

Step 1: Click on "Microsoft Excel" or "Text File"

On the Start Page, under Connect, you will see options like:

Microsoft Excel (for .xlsx)

Text File (for .csv)

Click the correct option.

Step 2: Select Your File

Browse your computer.

Select your dataset file.

Click Open.

Step 3: Data Source Page

Your sheet names will appear on the left side.

Drag the sheet to the centre area.

Tableau will show a preview of your data.

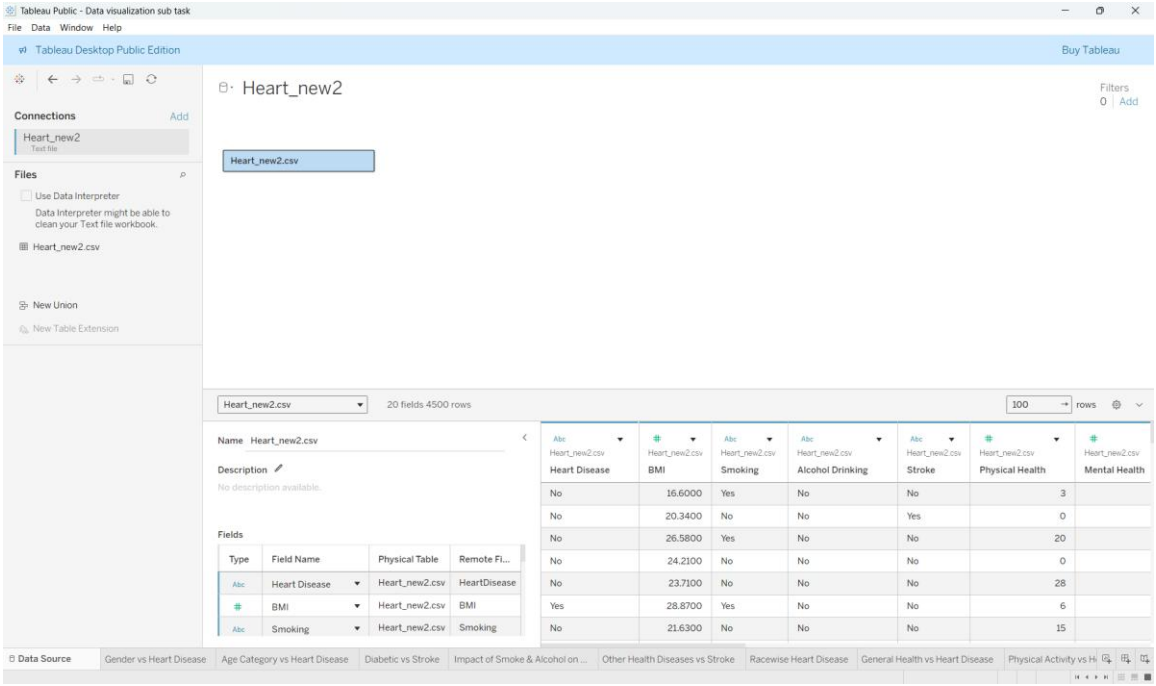


Tableau Public - Data visualization sub task
File Data Window Help
Tableau Desktop Public Edition
Buy Tableau
Connections
Add
Heart_new2
Text file
Files
Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.
Heart_new2.csv
New Union
New Table Extension
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Name Heart_new2.csv
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Fields
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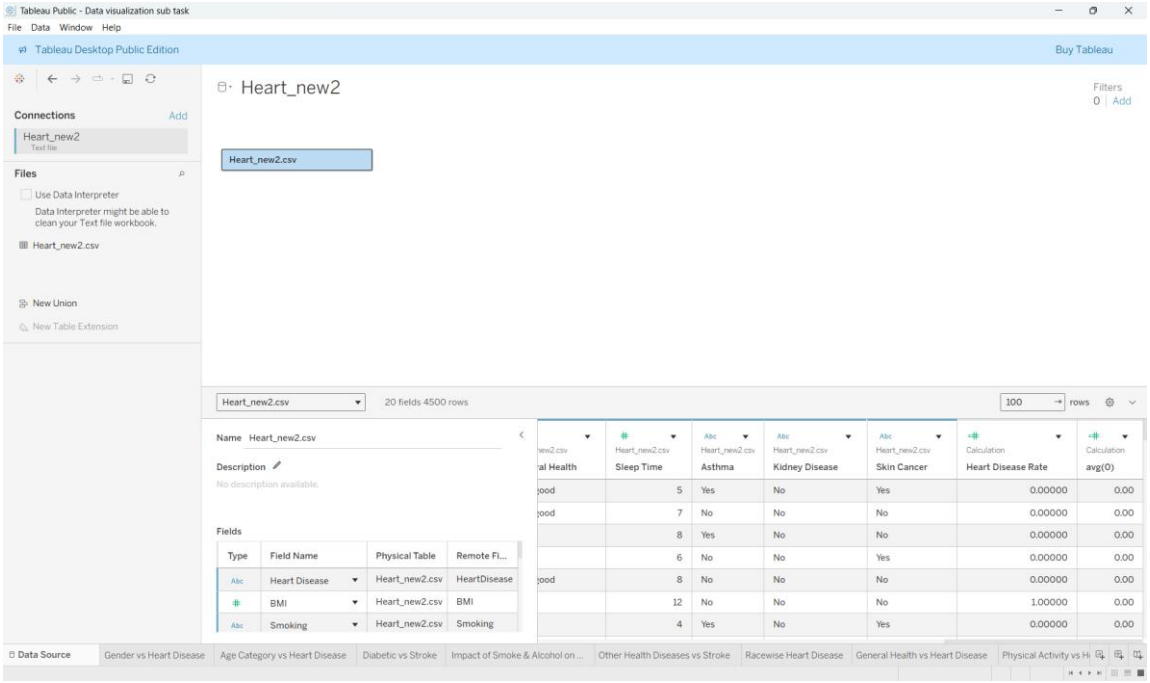
Heart Disease Heart_new2.csv HeartDisease
BMI Heart_new2.csv BMI
Smoking Heart_new2.csv Smoking
Heart Disease BMI Smoking Alcohol Drinking Stroke Physical Health Mental Health
No 16.6000 Yes No No 3
No 20.3400 No No Yes 0
No 26.5800 Yes No No 20
No 24.2100 No No No 0
No 23.7100 No No No 28
Yes 28.8700 Yes No No 6
No 21.6300 No No No 15
Data Source
Gender vs Heart Disease
Age Category vs Heart Disease
Diabetic vs Stroke
Impact of Smoke & Alcohol on ...
Other Health Diseases vs Stroke
Racewise Heart Disease
General Health vs Heart Disease

2) Data Preparation:

❖ Data Cleaning

Preparing the data for visualization involves cleaning the data to remove irrelevant or missing data, transforming the data into a format that can be easily visualized, exploring the data to identify patterns and trends, filtering the data to focus on specific subsets of data, preparing the data for visualization software, and ensuring the data is accurate and complete.



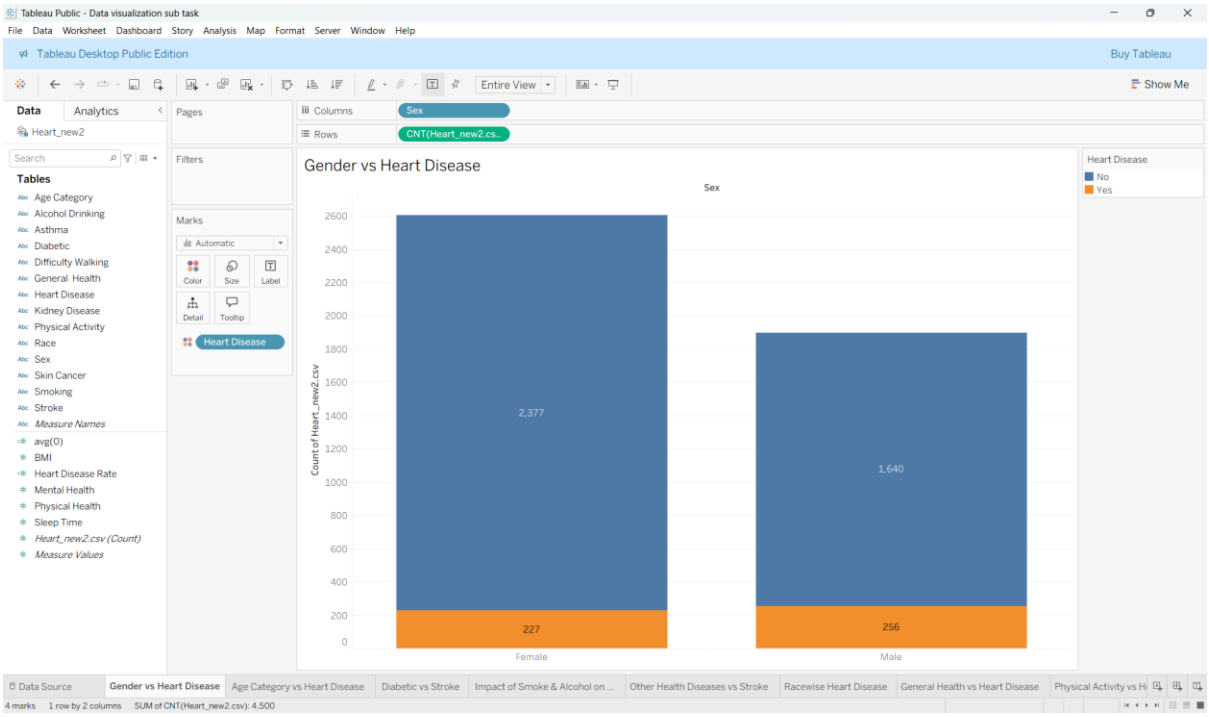
3) Data Visualization:

Data visualization is the process of creating graphical representations of data to help people understand and explore the information. The goal of data visualization is to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

❖ No of Unique Visualizations:

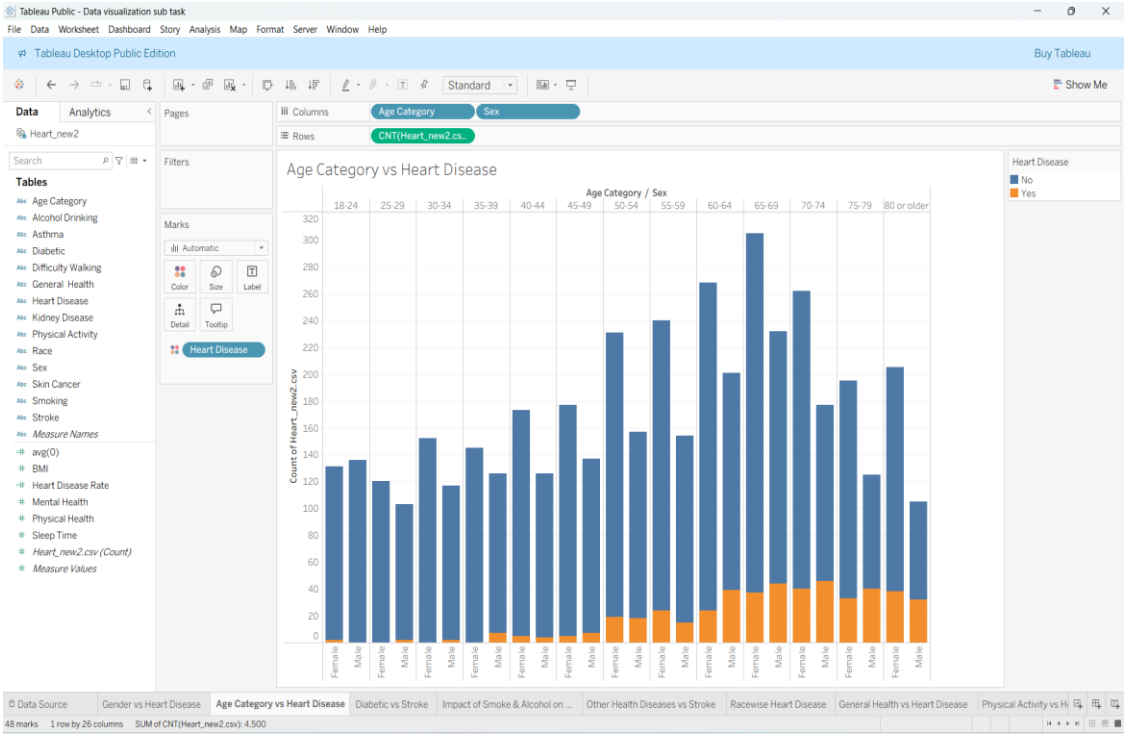
1) Gender Vs Heart Disease :

This visualization shows the relationship between **gender and the occurrence of heart disease**. It compares the number of heart disease cases among **male and female individuals**. The chart helps identify which gender is more affected by heart disease in the dataset.



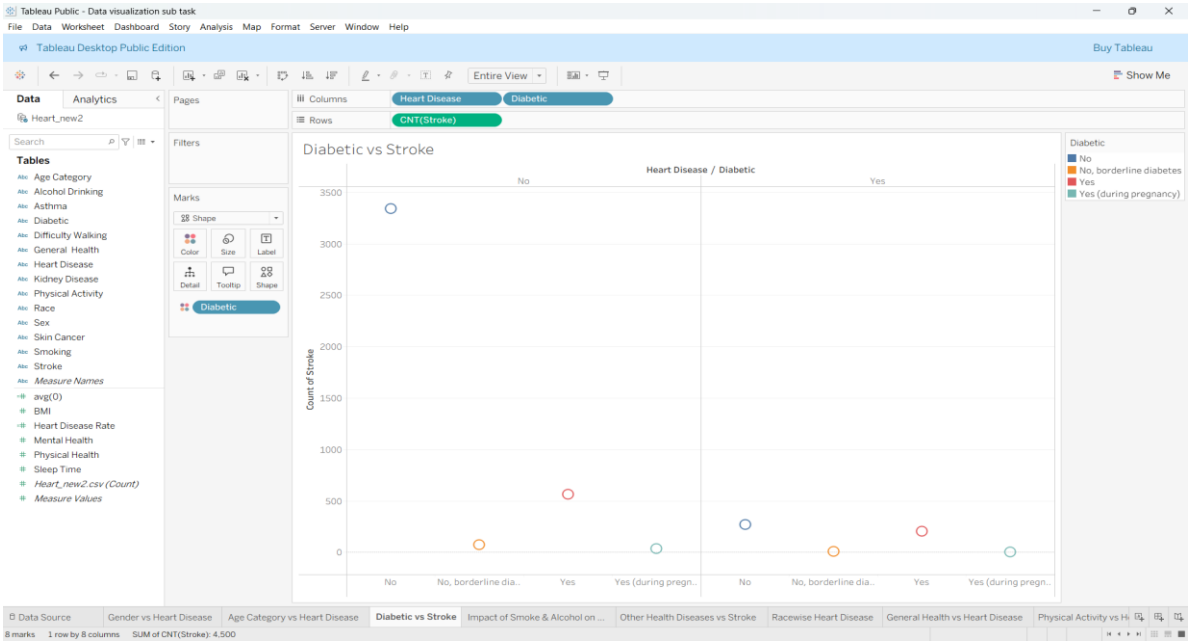
2) Age Category vs Heart Disease

This visualization analyzes how **heart disease cases vary across different age groups**. It helps identify whether the risk of heart disease increases with age. The chart clearly shows which age category is more vulnerable to heart disease.



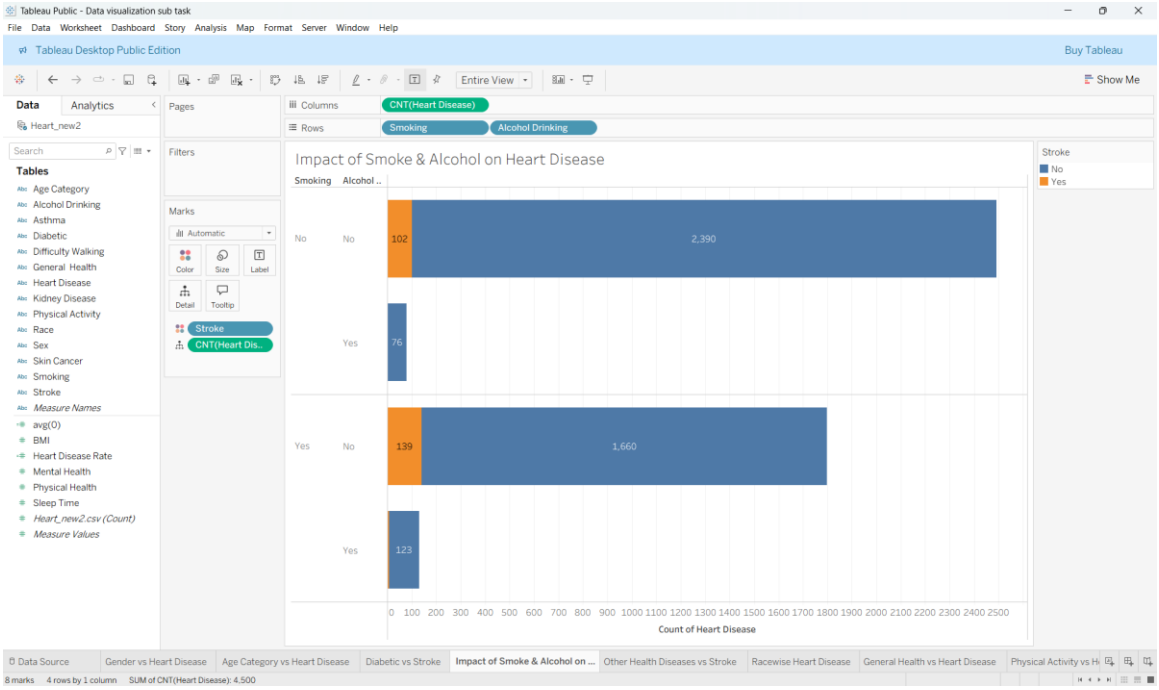
3) Diabetic vs Stroke

This visualization highlights the relationship between **diabetes and stroke occurrences**. It helps understand whether people suffering from diabetes have a higher chance of experiencing a stroke. The comparison provides insights into how diabetes can impact cardiovascular health.



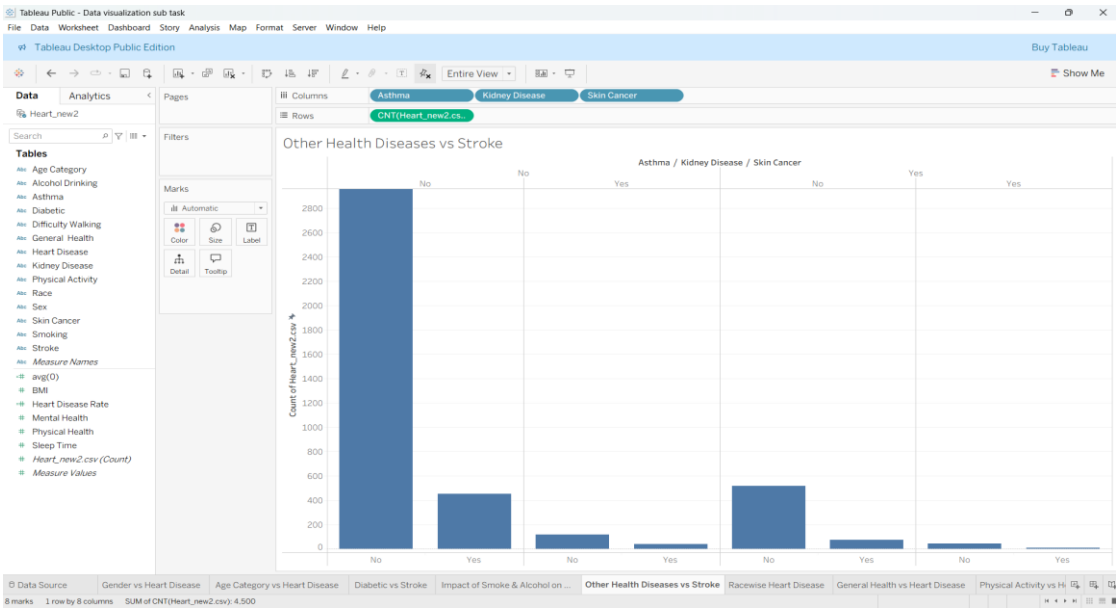
4) Impact of Smoke & Alcohol on Heart Disease

This chart examines how **smoking and alcohol consumption influence heart disease risk**. It compares individuals who smoke or drink alcohol with those who do not. The visualization helps identify lifestyle habits that increase the likelihood of heart disease.



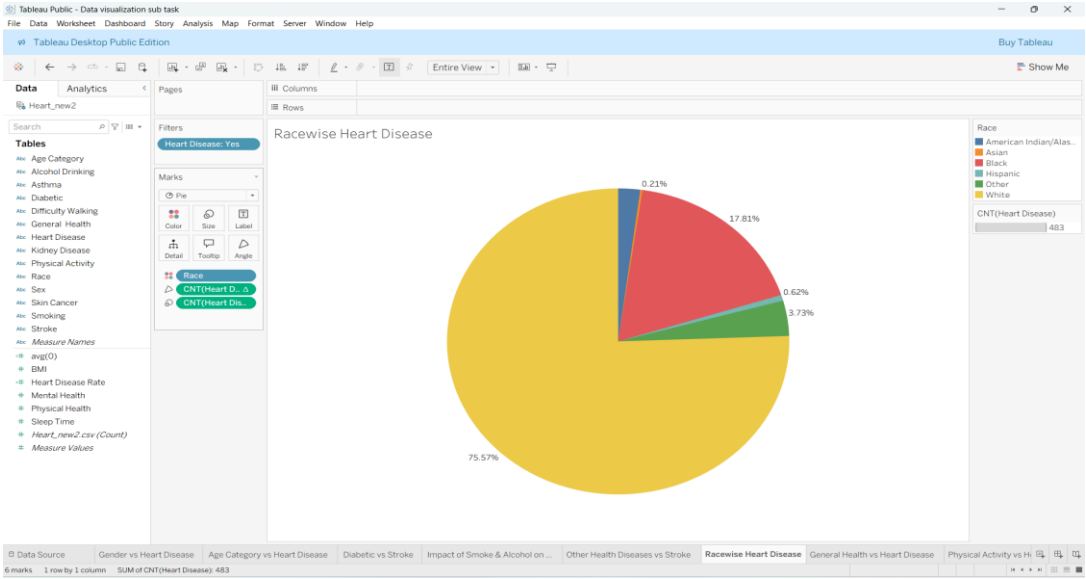
5) Other Health Diseases vs Stroke

This visualization studies the connection between **other health conditions such as asthma, kidney disease, or skin cancer and stroke cases**. It helps identify whether people with additional health issues are more likely to experience a stroke.



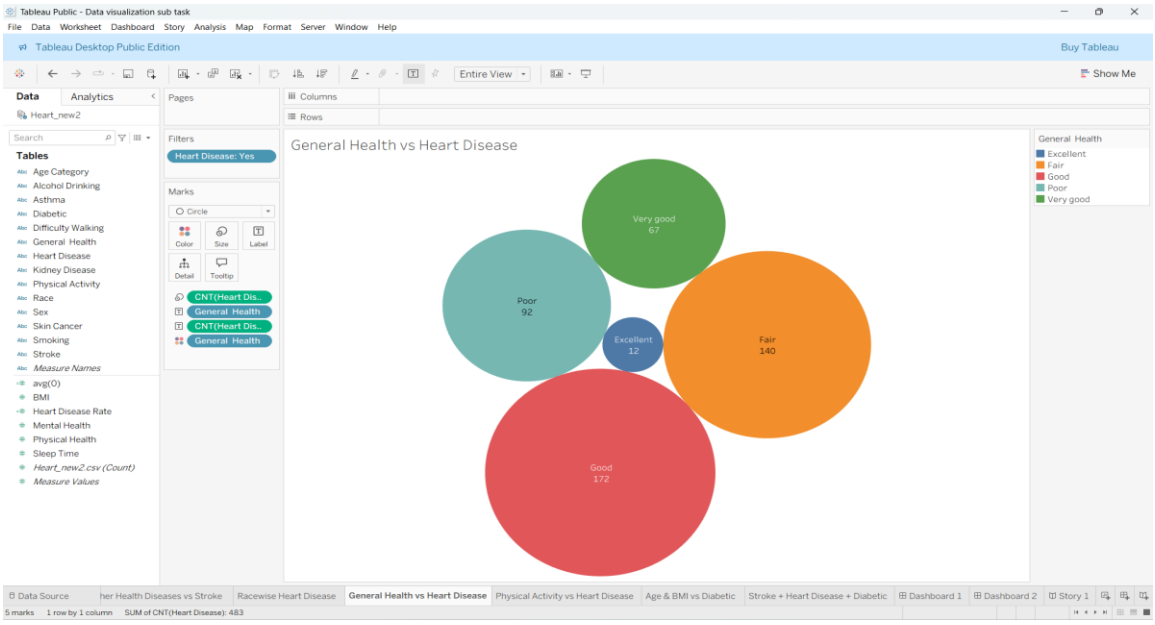
6) Racewise Heart Disease

This chart analyzes the distribution of **heart disease among different racial groups** in the dataset. It helps highlight how heart disease prevalence may vary among different populations. The visualization provides insights into demographic health patterns.



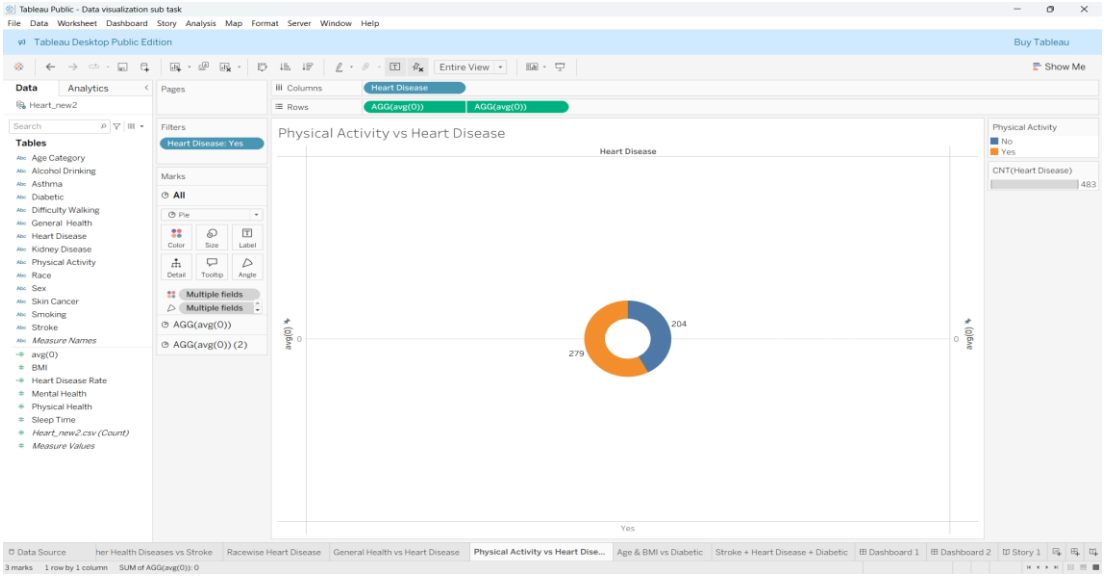
7) General Health vs Heart Disease

This visualization shows the relationship between **overall health condition and heart disease occurrence**. Individuals with poor or fair health are compared with those reporting good or excellent health. It helps understand how general health status affects heart disease risk.



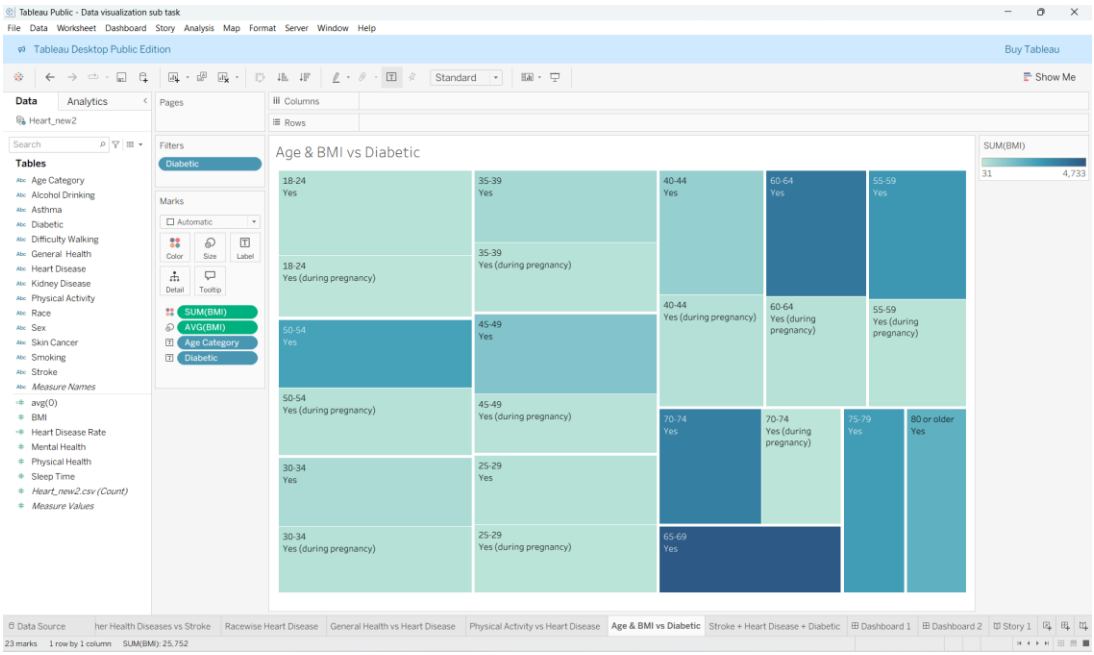
8) Physical Activity vs Heart Disease

This chart compares people who **engage in physical activity with those who do not**. It helps analyze how an active lifestyle may reduce the chances of heart disease. The visualization highlights the importance of physical exercise for heart health.



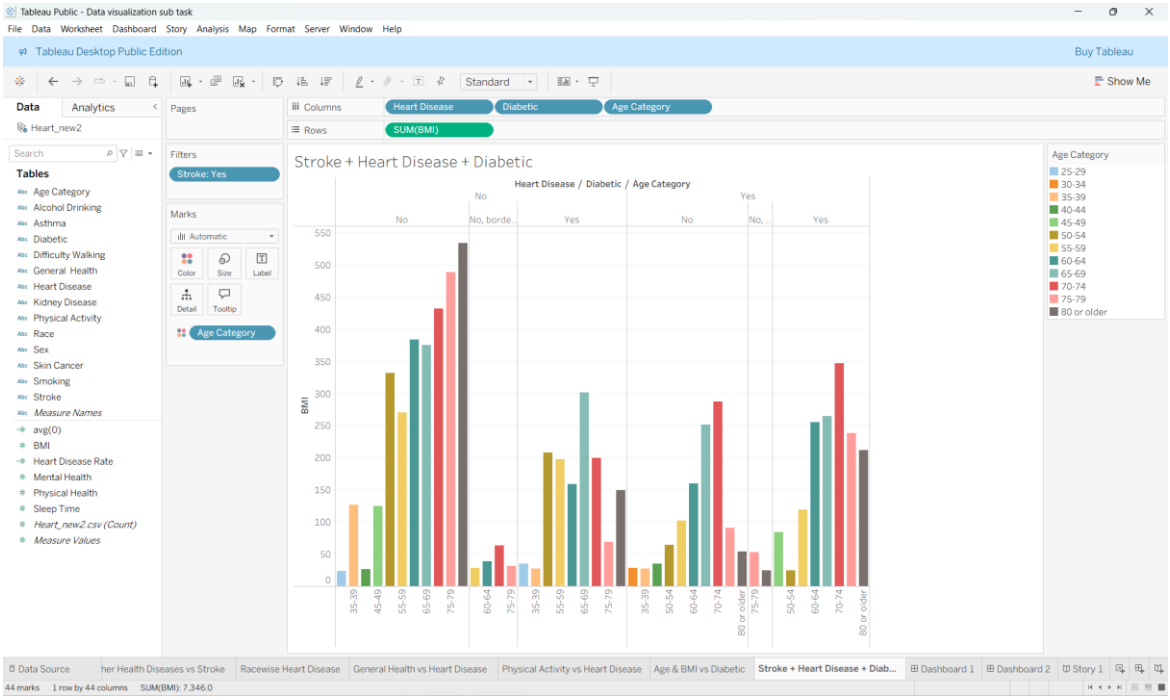
9) Age & BMI vs Diabetic

This visualization explores how **age and Body Mass Index (BMI) relate to diabetes cases**. It helps determine whether higher BMI and increasing age contribute to the risk of diabetes. The chart provides insight into health factors affecting diabetes prevalence.



10) Stroke + Heart Disease + Diabetic

This visualization examines individuals suffering from **multiple health conditions such as stroke, heart disease, and diabetes**. It highlights how these diseases are interconnected and may occur together in certain patients. The analysis helps understand the combined impact of these major health risks.



4) Dashboard

A dashboard is a graphical user interface (GUI) that displays information and data in an organized, easy-to-read format. Dashboards are often used to provide real-time monitoring and analysis of data and are typically designed for a specific purpose or use case. Dashboards can be used in a variety of settings, such as business, finance, manufacturing, healthcare, and many other industries. They can be used to track key performance indicators (KPIs), monitor performance metrics, and display data in the form of charts, graphs, and tables.

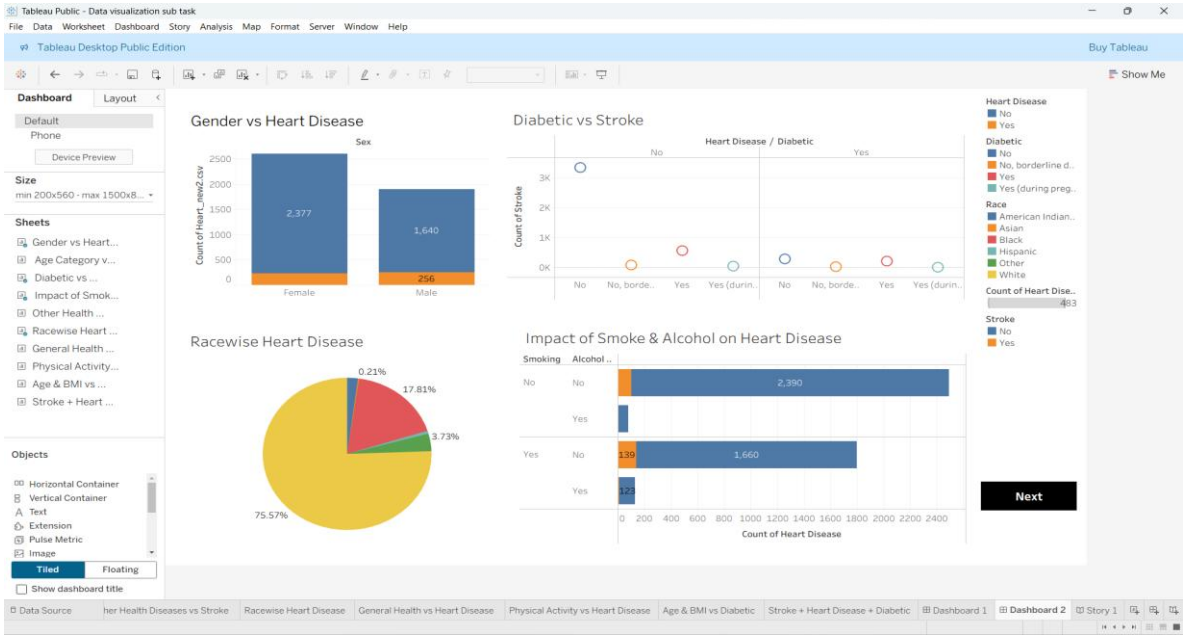
❖ Responsive and Design of Dashboard

Dashboard 1: Heart Disease Risk Factors Overview

This dashboard provides an overview of the major factors related to heart disease such as gender, race, smoking habits, alcohol consumption, and diabetes. The visualizations help identify patterns and relationships between lifestyle habits and heart disease occurrence. It allows users to easily understand how different demographic and behavioral factors influence heart disease risk.

Key Insights:

- Comparison of heart disease cases between male and female populations.
- Analysis of diabetes and stroke relationships.
- Visualization of smoking and alcohol impact on heart disease.
- Race-wise distribution of heart disease cases.

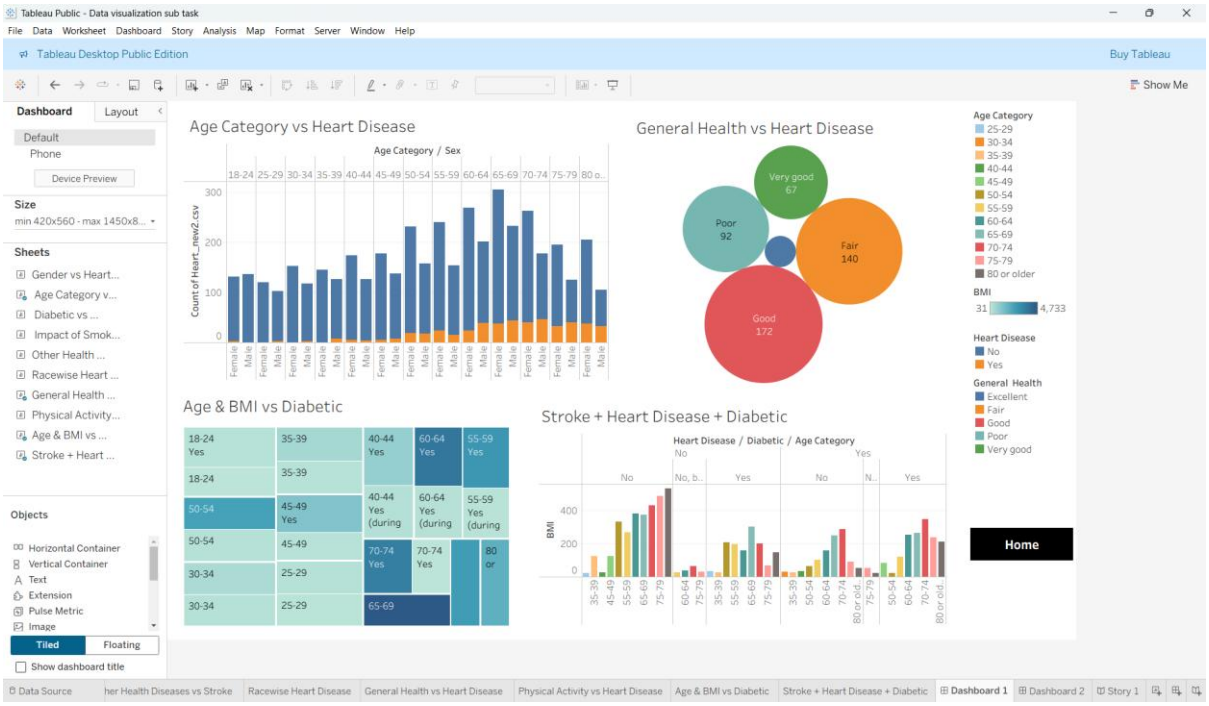


Dashboard 2: Health Conditions and Age Analysis

This dashboard focuses on how age, general health condition, BMI, and physical activity affect heart disease and related health conditions. It helps in understanding how age groups and overall health status influence the likelihood of heart disease, stroke, and diabetes.

Key Insights:

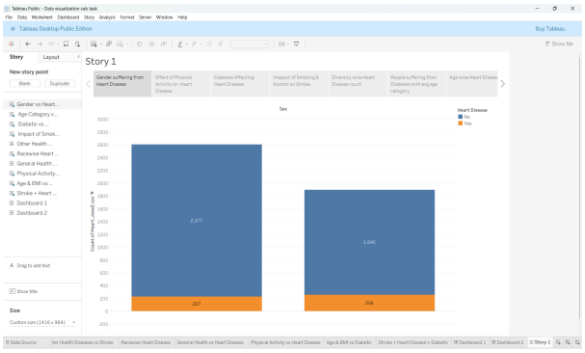
- Age category analysis showing how heart disease risk changes across different age groups.
- Relationship between general health condition and heart disease.
- Visualization of BMI and age impact on diabetic patients.
- Combined analysis of stroke, heart disease, and diabetes.

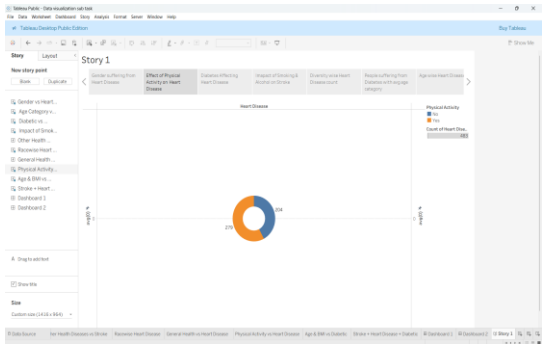


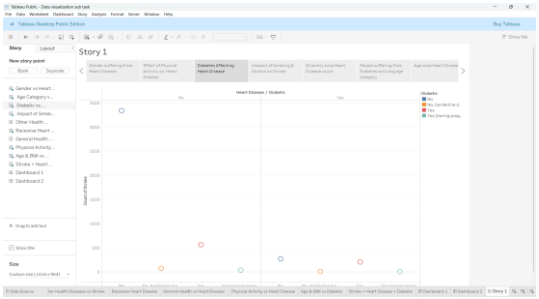
5) Storyboard

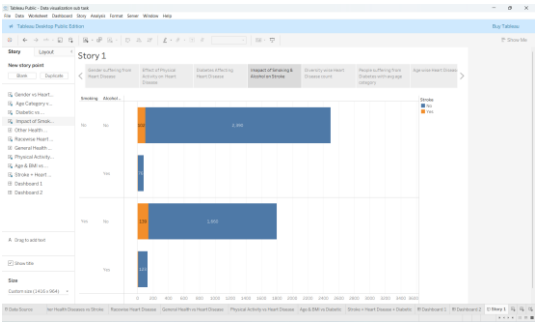
The number of scenes in a storyboard for a data visualization analysis of the Heart disease will depend on the complexity of the analysis and the specific insights that are trying to be conveyed. A storyboard is a visual representation of the data analysis process and it breaks down the analysis into a series of steps or scenes.

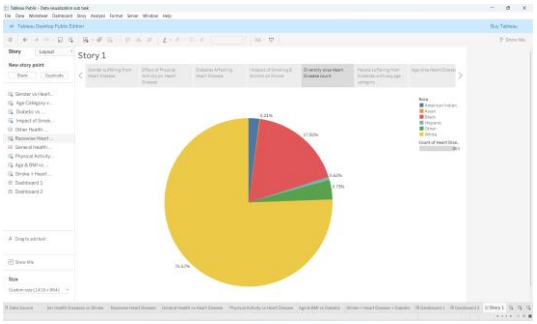
❖ Design a Storyboard

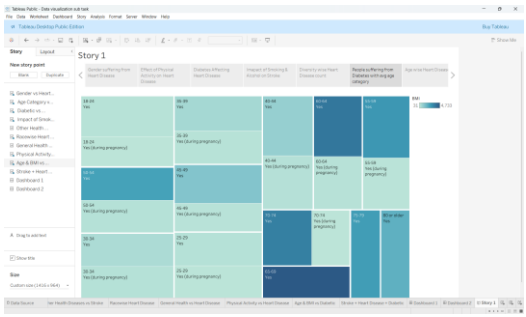


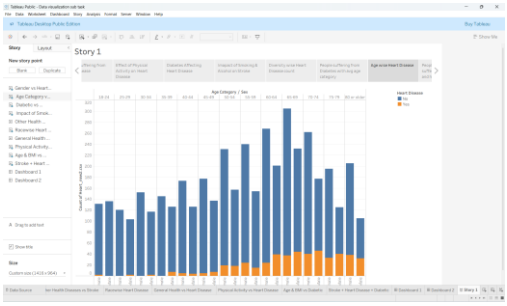


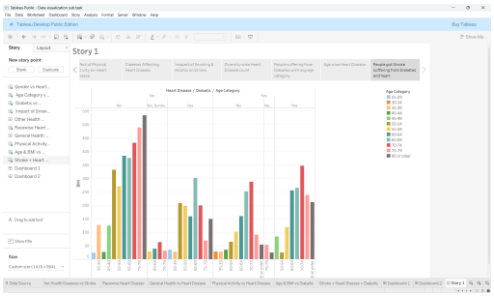










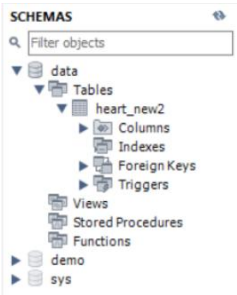


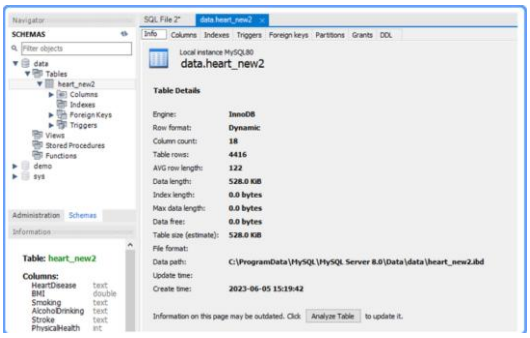
6) Performance Testing

Performance testing involves assessing the volume of data rendered from the database, the impact of data filters on system responsiveness, and the complexity introduced by the number of visualizations. Optimizing these factors ensures the dashboard operates efficiently, providing timely and reliable insights.

❖ Amount of Data DB

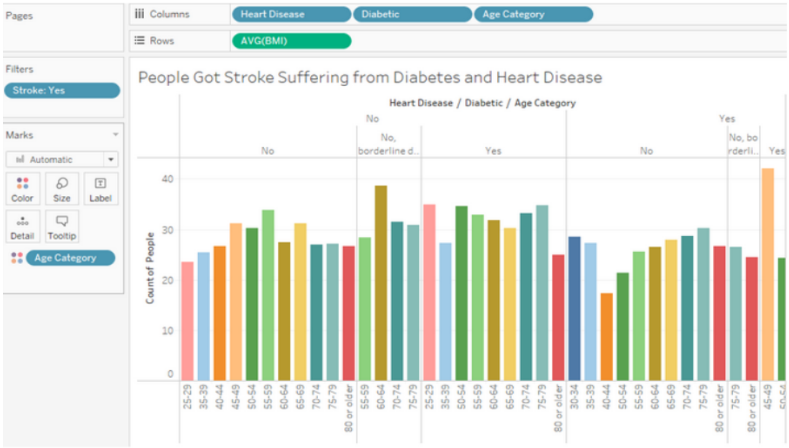
Monitor the volume of data being pulled and rendered from the database to ensure queries are optimized and not overloading the system.
The amount of data that is rendered to a database depends on the size of the dataset and the capacity of the database to store and retrieve data.
Open the MySQL Workbench, go to the database then click to expand the tables,select the table and click on (i) button to get the information related to table such as column count,table rows etc.

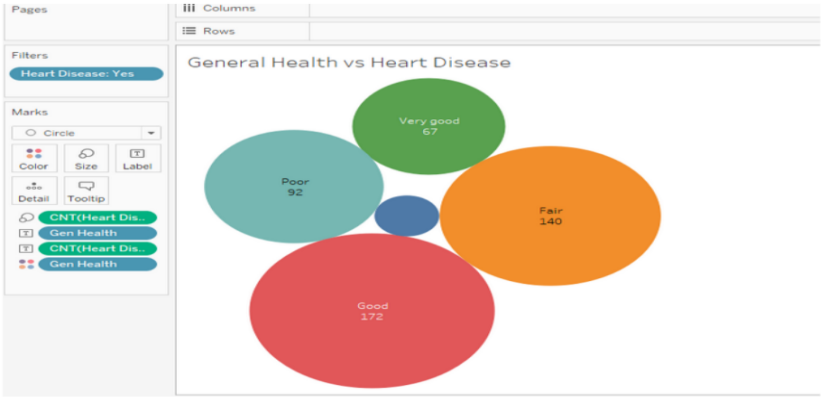




❖ Utilization of Filters

Utilization of data filters refers to the effective implementation and management of filtering mechanisms within the Project to refine and focus the dataset. Proper use of filters enhances performance by limiting the volume of data processed and displayed, thereby improving responsiveness. It also enables users to interactively explore specific segments of data, leading to more targeted and meaningful insights.





❖ No of Calculation Fields:

Calculated fields are used in **Tableau Desktop** to create new values from existing data. These fields help in better analysis and visualization of the dataset.

1. **Age Category** – Groups age into different categories for better comparison.
2. **BMI Category** – Classifies BMI values to analyze health conditions.
3. **Heart Disease Indicator** – Identifies whether a person has heart disease or not.
4. **Combined Health Condition** – Combines stroke, diabetes, and heart disease for analysis.
5. **Risk Factor** – Analyzes the impact of smoking, alcohol, and physical activity.

❖ No of Visualizations/ Graphs :

1. Gender wise Heart Disease
2. Age wise Heart Disease
3. People Suffering from Diabetic and Stroke
4. Impact of Smoking and alcohol drinking on heart disease
5. Other Diseases vs Stroke
6. Race wise Heart disease
7. General Health vs Heart Disease
8. Physical activity vs heart disease
9. Age and BMI vs Heart disease
10. People got stroke suffering from Diabetes and Heart disease

1) Web integration

Web integration of a Tableau involves embedding interactive visualizations into a website or web application. This allows users to explore data insights directly within a web interface, enhancing accessibility and engagement. It supports real-time updates, user filtering, and seamless navigation for a dynamic data storytelling experience.

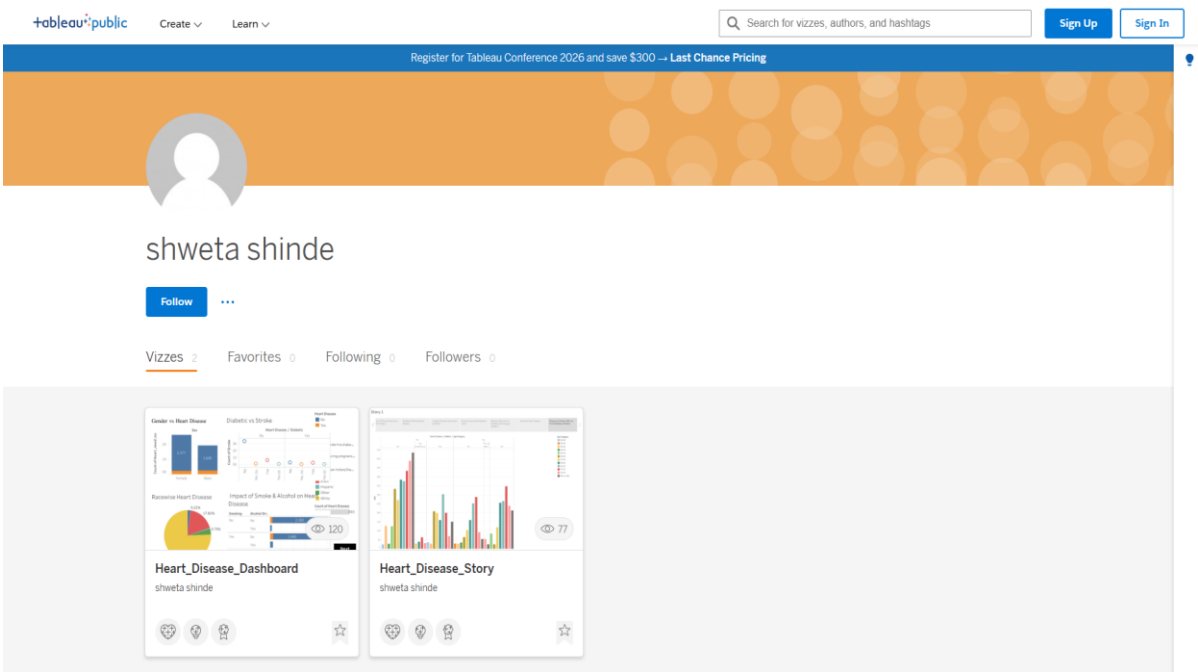
❖ Publishing

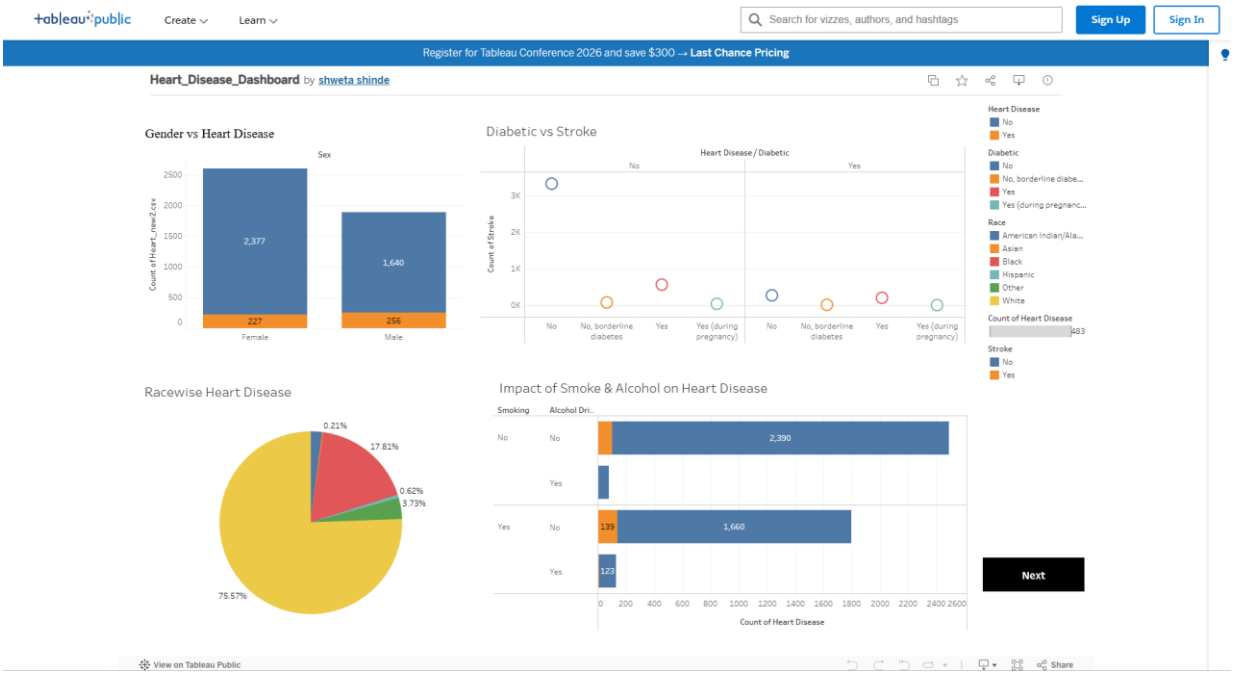
a) **Sign in to Tableau Public**

In Tableau Desktop, go to **File** > **Save to Tableau Public**.
If you're not already signed in, a login window will appear.
Enter your **Tableau Public credentials** or sign up if you don't have an account.

b) **Save and Publish**

- After logging in, you'll be prompted to **name your workbook**.
- Click **Save** – Tableau will upload the workbook to your Tableau Public profile.

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Sign Up
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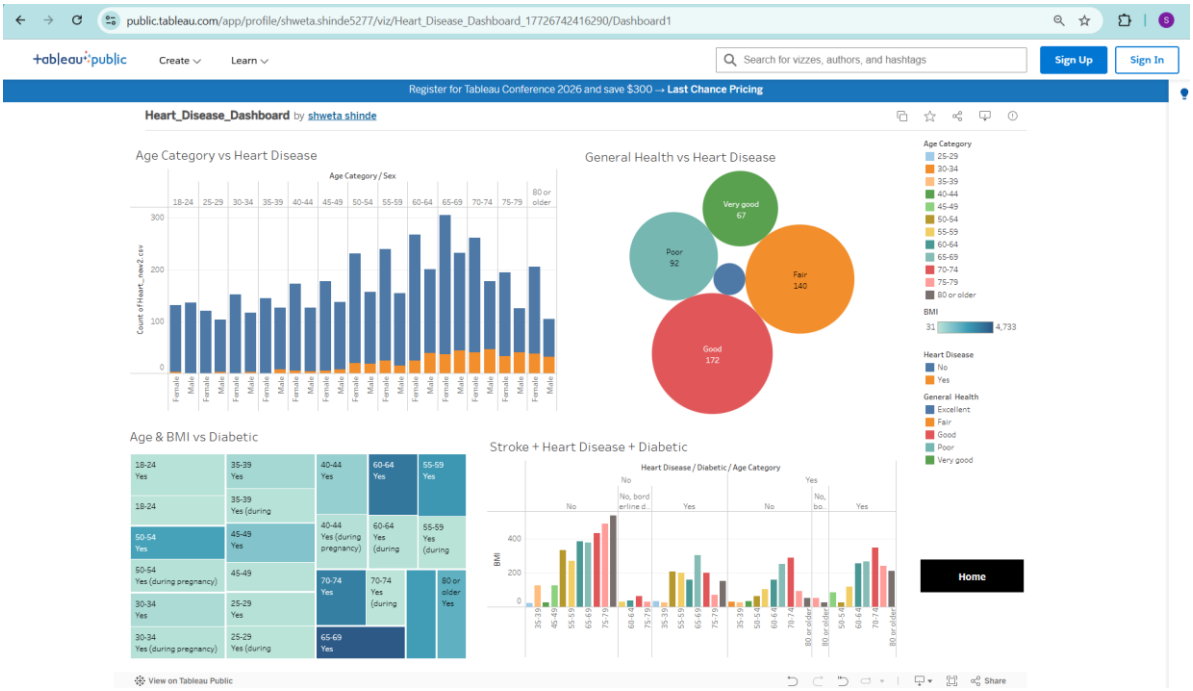
Heart_Disease_Dashboard by shweta shinde
Gender vs Heart Disease
Sex
2,377
1,640
227
256
Female
Male
Diabetic vs Stroke
Heart Disease / Diabetic
Count of Stroke
Heart Disease
Diabetic
Race
Count of Heart Disease
Stroke
Racewise Heart Disease
0.21%
17.81%
0.62%
3.73%
75.57%
Impact of Smoke & Alcohol on Heart Disease
Smoking
Alcohol Dri...
2,390
139
1,660
123
Count of Heart Disease
Next
View on Tableau Public
Share


public.tableau.com/app/profile/shweta.shinde5277/viz/Heart_Disease_Dashboard_17726742416290/Dashboard1
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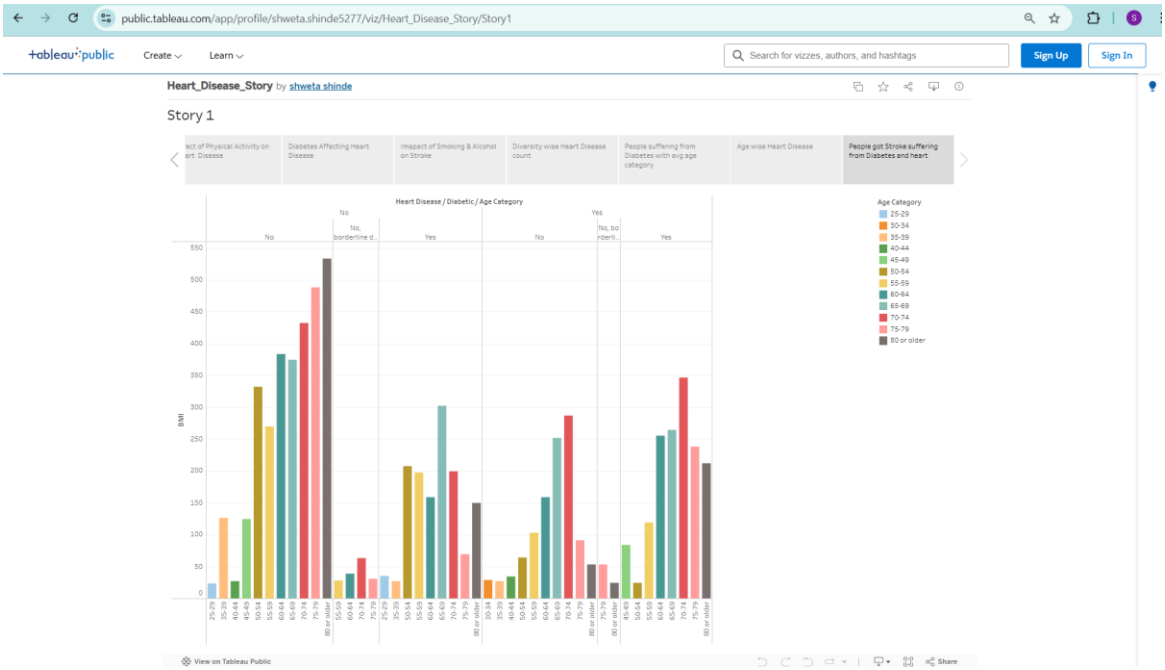
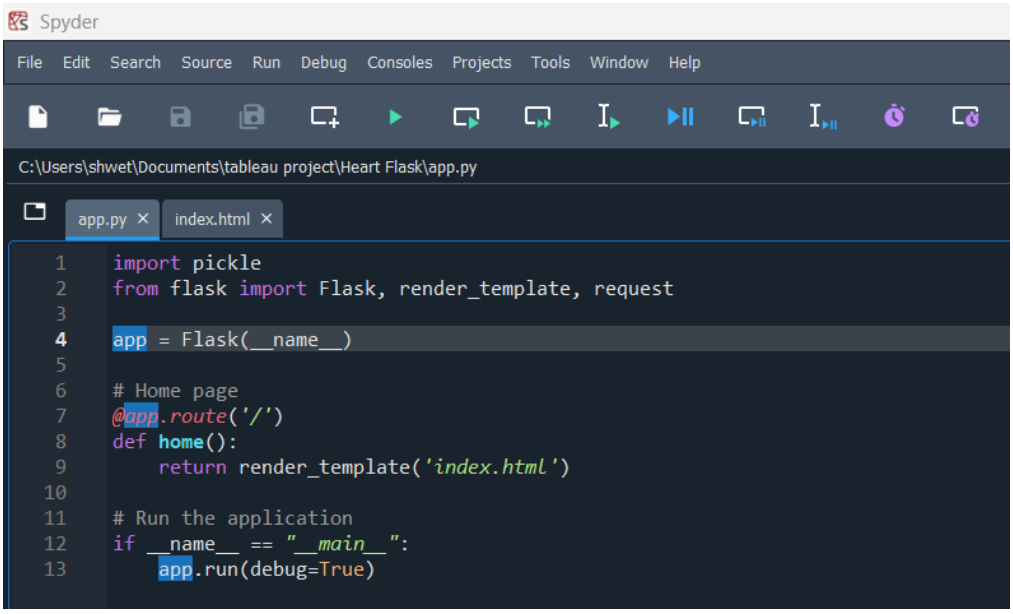
Heart_Disease_Dashboard by shweta shinde
Age Category vs Heart Disease
Age Category / Sex
General Health vs Heart Disease
Very good 67
Poor 92
Fair 140
Good 172
Age Category
BMI
31
4,733
Heart Disease
General Health
Excellent
Fair
Good
Poor
Very good
Age & BMI vs Diabetic
Stroke + Heart Disease + Diabetic
Heart Disease / Diabetic / Age Category
BMI
Home
View on Tableau Public
Share



❖ Dashboard and Story embed with UI with Flask.

In this activity, you will create a simple web application using Flask and embed a Tableau Dashboard or Story into the UI. This allows users to view and interact with Tableau visualizations directly through a custom web interface.

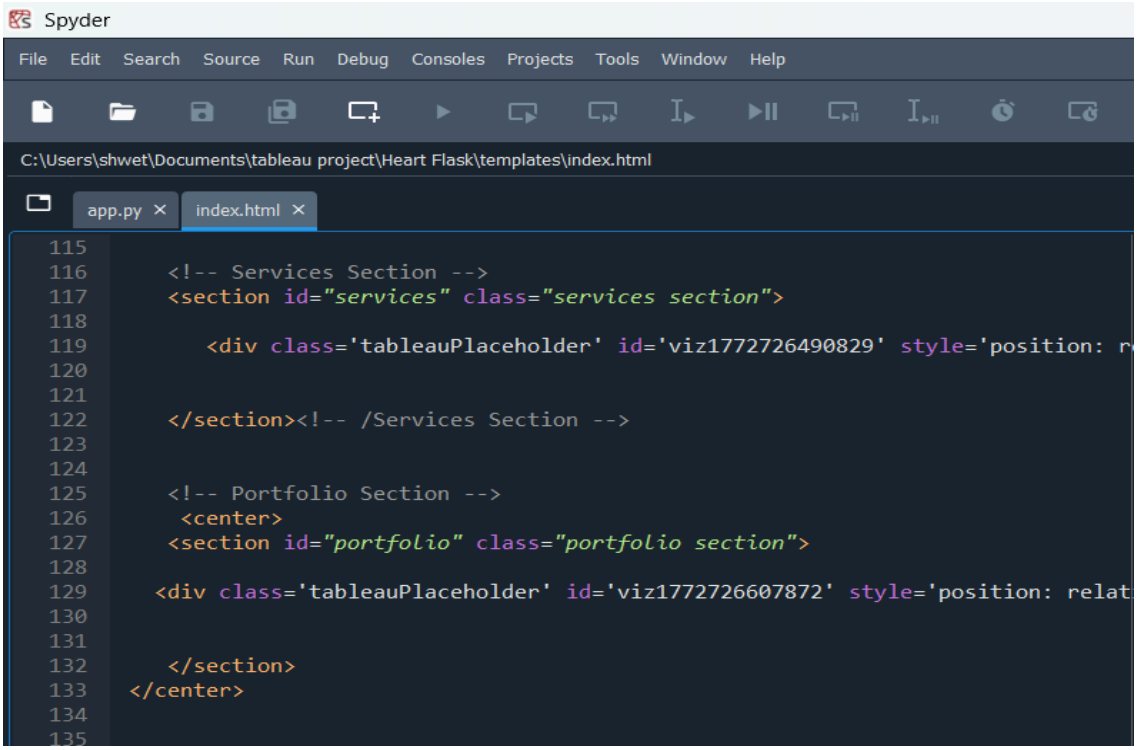
1) app.py



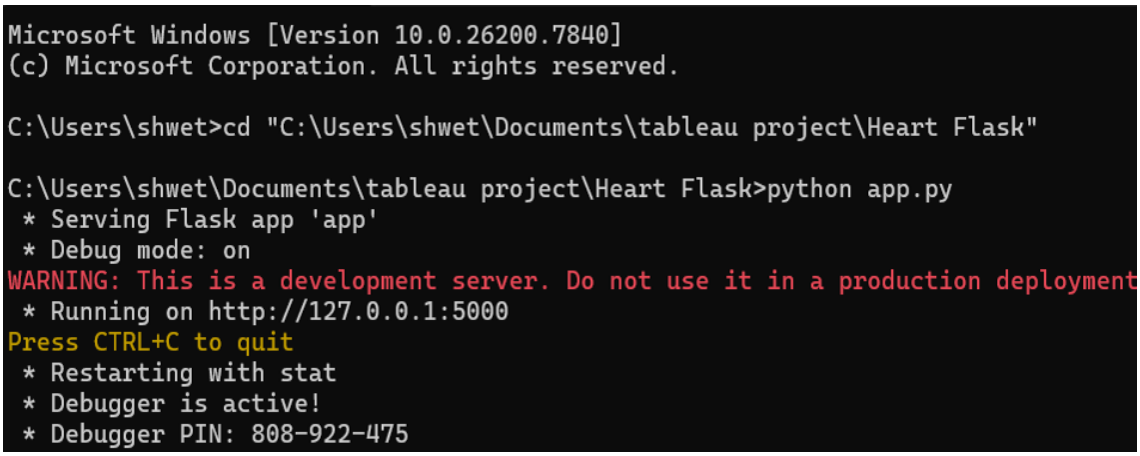

```python
import pickle
from flask import Flask, render_template, request

app = Flask(__name__)

# Home page
@app.route('/')
def home():
    return render_template('index.html')

# Run the application
if __name__ == "__main__":
    app.run(debug=True)
```

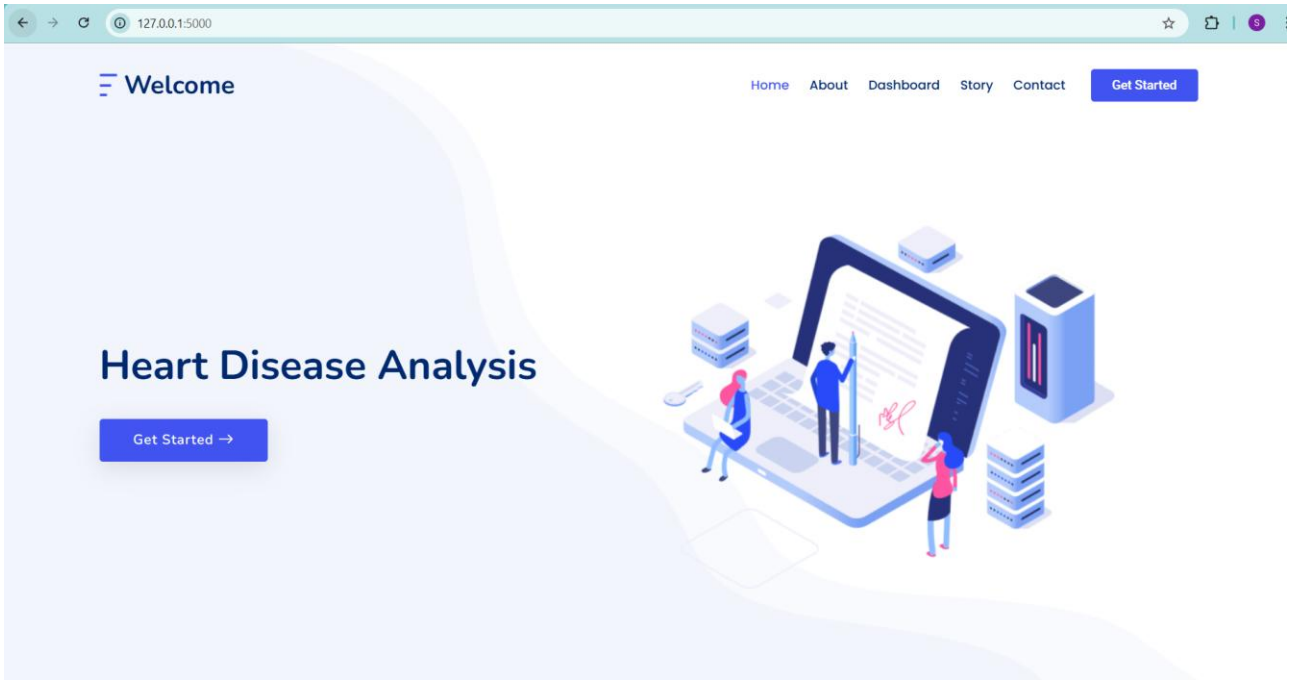
2) Adding Dashboard and Story on HTML page.

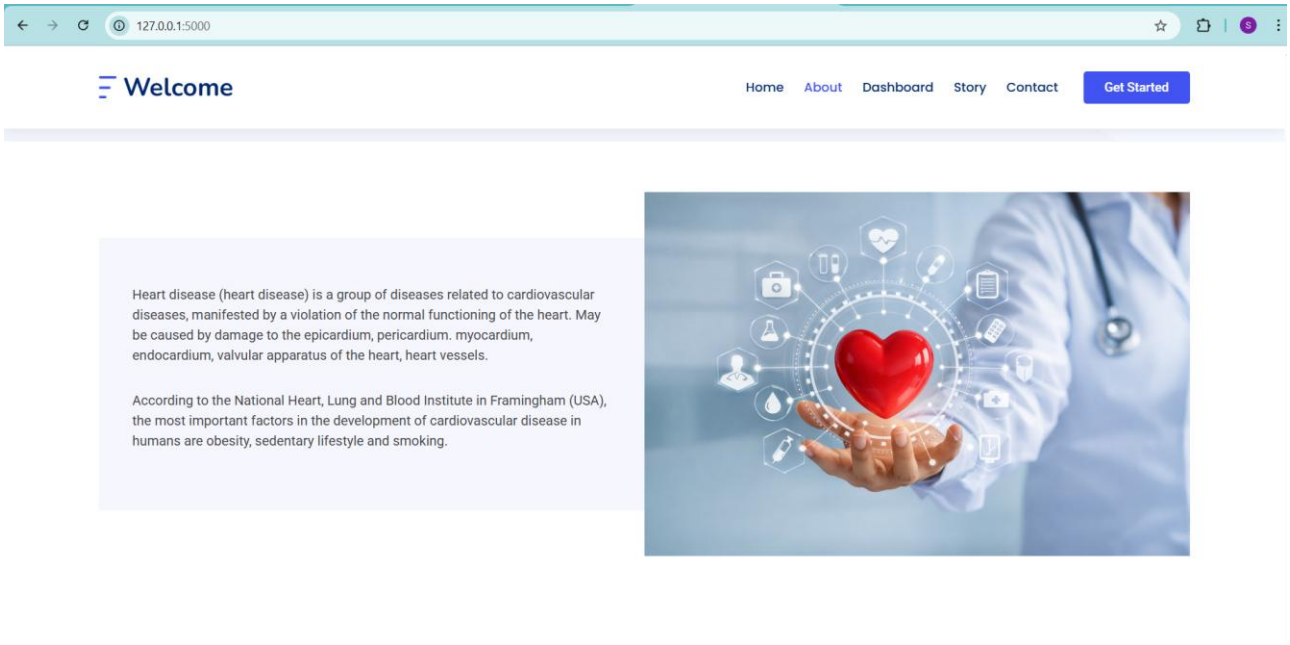
a) Open Tableau Public profile and open Dashboard/Story.
b) Click on share option and copy the embed code.
c) Paste in HTML page

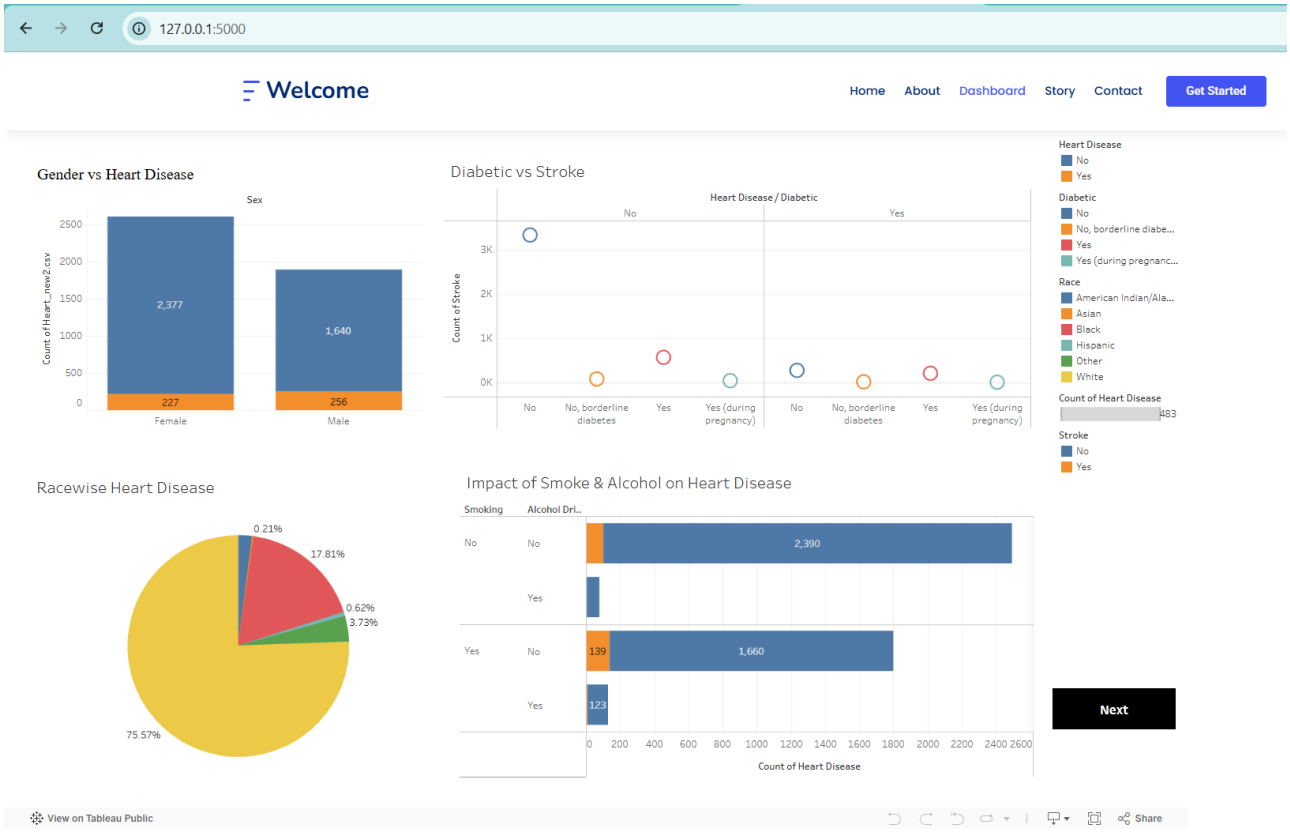


d) Run the app.py and copy the port number and paste it on Browser.

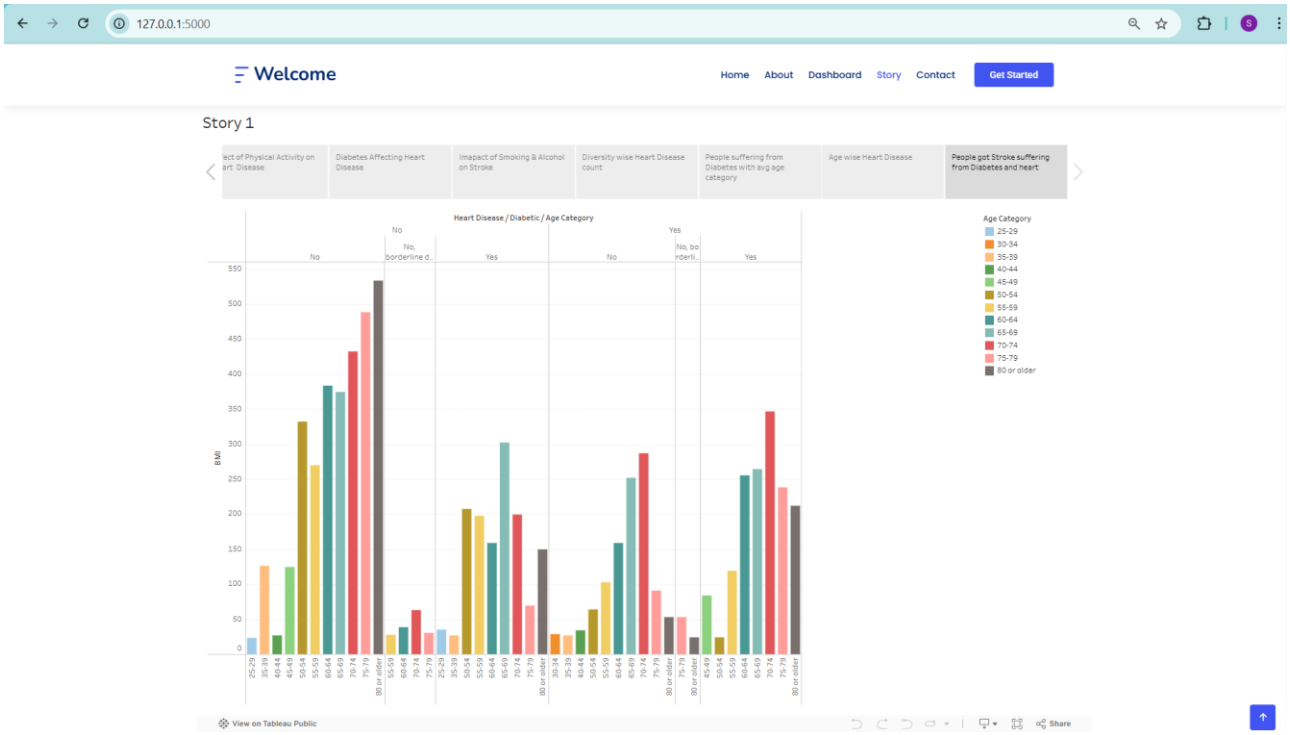


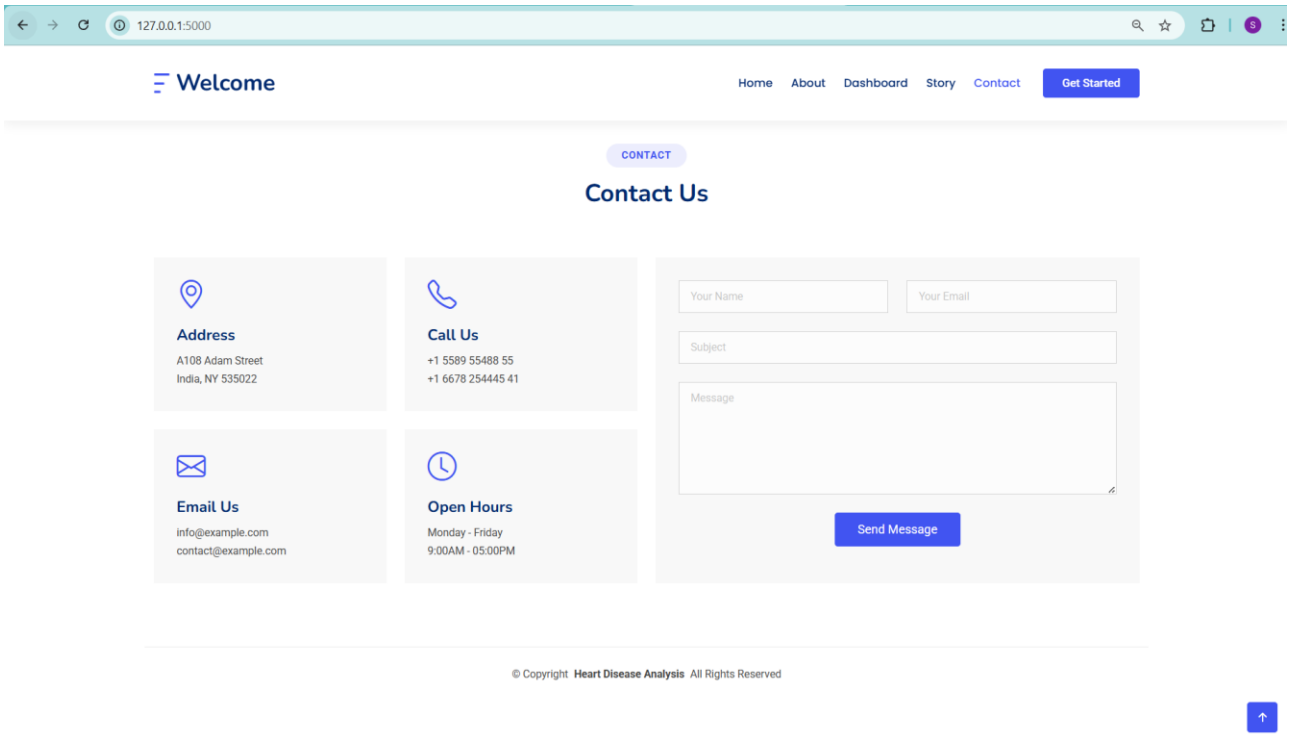

127.0.0.1:5000
Welcome
Home
About
Dashboard
Story
Contact
Get Started
Heart Disease Analysis
Get Started →


127.0.0.1:5000
Welcome
Home
About
Dashboard
Story
Contact
Get Started
Heart disease (heart disease) is a group of diseases related to cardiovascular diseases, manifested by a violation of the normal functioning of the heart. May be caused by damage to the epicardium, pericardium. myocardium, endocardium, valvular apparatus of the heart, heart vessels.
According to the National Heart, Lung and Blood Institute in Framingham (USA), the most important factors in the development of cardiovascular disease in humans are obesity, sedentary lifestyle and smoking.


127.0.0.1:5000
Welcome
Home
About
Dashboard
Story
Contact
Get Started
Gender vs Heart Disease
Sex
Count of Heart_new2.csv
2,377
1,640
227
256
Female
Male
Diabetic vs Stroke
Heart Disease / Diabetic
No
Yes
Count of Stroke
No
No, borderline diabetes
Yes
Yes (during pregnancy)
Heart Disease
No
Yes
Diabetic
No
No, borderline diabe...
Yes
Yes (during pregnanc...
Race
American Indian/Ala...
Asian
Black
Hispanic
Other
White
Count of Heart Disease
483
Stroke
No
Yes
Racewise Heart Disease
0.21%
17.81%
0.62%
3.73%
75.57%
Impact of Smoke & Alcohol on Heart Disease
Smoking
Alcohol Dri..
2,390
139
1,660
123
Count of Heart Disease
Next
View on Tableau Public
Share

127.0.0.1:5000
Welcome
Home
About
Dashboard
Story
Contact
Get Started
Age Category vs Heart Disease
Age Category / Sex
18-24
25-29
30-34
35-39
40-44
45-49
50-54
55-59
60-64
65-69
70-74
75-79
80 or older
Count of Heart_new2.csv
General Health vs Heart Disease
Very good
67
Poor
92
Fair
140
Good
172
Age Category
25-29
30-34
35-39
40-44
45-49
50-54
55-59
60-64
65-69
70-74
75-79
80 or older
BMI
31
4,733
Heart Disease
No
Yes
General Health
Excellent
Fair
Good
Poor
Very good
Age & BMI vs Diabetic
Stroke + Heart Disease + Diabetic
Heart Disease / Diabetic / Age Category
BMI
Home
View on Tableau Public
Share


127.0.0.1:5000
Welcome
Home
About
Dashboard
Story
Contact
Get Started
Story 1
Diabetes Affecting Heart Disease
Imapact of Smoking & Alcohol on Stroke
Diversity wise Heart Disease count
People suffering from Diabetes with avg age category
Age wise Heart Disease
People got Stroke suffering from Diabetes and heart
Heart Disease / Diabetic / Age Category
Age Category
25-29
30-34
35-39
40-44
45-49
50-54
55-59
60-64
65-69
70-74
75-79
80 or older
BMI
View on Tableau Public
Share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Welcome
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Get Started
CONTACT
Contact Us
Address
A108 Adam Street
India, NY 535022
Call Us
+1 5589 55488 55
+1 6678 254445 41
Email Us
info@example.com
contact@example.com
Open Hours
Monday - Friday
9:00AM - 05:00PM
Your Name
Your Email
Subject
Message
Send Message
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